

From symbolism to information? – Decoding the gene code

FRODE KJOSAVIK

Department of Economics and Resource Management (IØR), Norwegian University of Life Sciences (UMB), P.O. Box 5003, N-1432, Aas, Norway (e-mail: frode.kjosavik@umb.no; phone: +47-64-965672; fax: +47-64-965701)

Received 1 November 2005; accepted in revised form 22 August 2006

Key words: Arbitrariness, Causal information, Code, Functions, Intentionality, Semantic content, Symbols

Abstract. ‘Information’ and ‘code’ originated as technical terms within linguistics and information theory but are now widely used in genetics and developmental biology. Against this background, it is examined if coded information distinguishes genes from other information carriers, i.e., whether there are genetic words or sentences by virtue of the genetic code, and, if so, whether they have any semantic content. It is concluded that there is no genetic language with semantic content, but that the genetic code still enables unique language-like modes of transmission and interpretation of causal information.

Introduction

The informational role of genes in development and evolution is a central issue within the philosophy of biology.¹ After the discovery of epigenetic systems of inheritance, genes can no longer be taken to be unique simply because they carry information across generations. Crucial questions are rather whether genes differ from other causal factors with regard to *how* they carry information or the *kind* of information they carry. The abundance of metaphors from linguistics and informatics suggests an affirmative answer to both these questions, in that genes apparently make up a symbolic information system – a genetic ‘code’ – which resembles a Morse code – and thereby also carry a peculiar kind of information – namely, one that is *semantic*, as opposed to purely causal.

To decide if genes stand out from other information carriers, it is therefore important to establish whether or not there is a genetic code in the first place, and, if so, just how similar this is to a linguistic code. In this paper, I shall argue that there is indeed a genetic code, but only for proteins, and not at the higher level of phenotypes, and I shall examine if the resemblance between the genetic code and a linguistic code suffices to distinguish genes from non-coding causal

¹Cf., e.g., Mahner and Bunge (1997), Sterelny and Griffiths (1999), Maynard Smith (2000a, b), Sterelny (2000), Godfrey-Smith (2000b), Sarkar (2000), Winnie (2000), Gray et al. (2001), Jablonka (2001, 2002), Moss (2004) and Stegmann (2004).

factors that affect development and evolution. If the code makes up a symbolic system of information, genes would be privileged in terms of combinatorial potential, and thus in a weak way. If the code in addition has semantic content, like the Morse code, that would set genes apart from other non-coding causal factors in the same strong way as words or sentences that have semantic content in linguistic communication.

By way of an analogy with the Morse code, I shall propose a strong criterion of semantic content, namely, arbitrariness and aboutness construed so as to resemble properties of a man-made code-script. The strong criterion is motivated by the fact that we are looking for a condition which distinguishes genes in a language-like manner from other causal factors in terms of what kind of information they carry. As the same criterion of artificial code resemblance seems to underlie the idea that there is a genetic program in organisms, and that causal factors in general are not instructions, my analysis will also be relevant to assess whether this computer analogy is warranted or not.

On the basis of this strong semantic criterion, I shall argue that the genetic information system does not have semantic content, unlike a linguistic code. This is taken to be equivalent to the claim that there can be no ascription of semantic content to a gene type. Moreover, if semantic content were to be ascribed only to a proper subset of gene tokens, or if the semantic criterion is weakened to mere arbitrariness – so that it would apply even to gene types – we no longer have a theory of semantic information that supports it.

If, on the other hand, the criterion is weakened to mere aboutness, which is backed by a ‘teleosemantic’ theory of information, it would not apply to gene types, and gene tokens would only fulfil it on a par with everything else for which there has been natural selection. It follows that the so-called *parity thesis*,² according to which causal information fails to single out genes from other information carriers, can be extended to semantic information as well. The latter is consistent with ‘Developmental Systems Theory’,³ but it does not imply it. Hence, genes can at most be distinguished from non-coding causes in the weak way, namely, with regard to how they carry their information – not because of the kind of information they carry.

The very notion of a code

A code in the relevant sense can be taken to be either a combinatorial system or a correspondence between two such systems. Thus, one-stranded DNA is based on a quaternary combinatorial code where the ‘letters’ are two purine bases – adenine and guanine – and two pyrimidine bases – thymine and cytosine. Two-stranded DNA is based on a correspondence code, or a mapping of a combinatorial code onto itself, namely, on the complementarity between the purine

²Cf. Griffiths and Gray (1994).

³Cf. Oyama (2000), Gray (1992) and Griffiths and Gray (1994).

and the pyrimidine bases, i.e., between adenine and thymine, and between guanine and cytosine, which is spelt out chemically through hydrogen bonding. The correspondence is in this case strict, or a one–one mapping, and because of the role it plays in the semi-conservative duplication of DNA, a molecular gene is often said ‘to code for’ its own replication. I shall call this the *replication code*.

In addition to the DNA system, there is the RNA system, which is like the former, except that uracil has been substituted for thymine. This means that there is also a correspondence code between the DNA and the RNA system – which I shall call the *transcription code* – as well as a correspondence between the RNA system and a protein system – the *translation code*. However, one has to distinguish between two different versions of the RNA system. There is the primary mRNA transcript, or pre-mRNA, which contains large regions that correspond to the *introns*, i.e., so-called non-coding regions within ‘coding’ regions of genes, and it turns out that these regions can be cut out in different ways, so as to yield different mature mRNA transcripts, corresponding to different sequences of amino acids.⁴ This – together with other complexities in the processes of transcription and translation – means that there can at most be a correspondence code between the DNA system and the primary RNA system, as well as between the mature RNA system and the protein system. There is no overall correspondence code between the DNA system and the protein system, though. So if one is to speak of a “genetic code”, and of “genes coding for proteins”, this can only be understood within an idealization that ignores the complexities of the process that leads from genes to assembled proteins. From now on, whenever we speak of “the genetic code”, we must bear this in mind. Given this idealization, the genetic code is then simply the transcription and the translation codes put together. Moreover, since one and the same amino acid may be encoded by different base triplets within the mature RNA system, there is of course no strict correspondence, or one–one mapping, even between the mature RNA system and the protein system, but only a many–one mapping. As long as the same base triplet cannot give rise to different amino acids, however, this many–one mapping may still count as a correspondence, if only in a degenerative sense.

Clearly, both the replication, transcription and translation codes have something in common with a man-made code, like the Morse code. Let us therefore pinpoint what are the features of such correspondence codes. In general, *x can be said to encode y* in the relevant sense when the following conditions obtain: (1) *x* and *y* are sequences of *combinatorial units*, i.e., characters or ‘letters’, from two sets – *X* and *Y* – the character lists or ‘alphabets’ (*combinatorial discreteness*) (2) there is a correlational function between minimal sequences of combinatorial units – the *correlational units* – from *X* and *Y* (*one–one or many–one correspondence*) (3) the position of a correlational unit within *x* is the same as the position of the correlational unit within *y* that it is

⁴See Lopez (1998).

mapped onto (*pattern congruence*); (4) x is either intended to signify y , or x – or a token of x – is a natural cause of y – or a token of y (*asymmetry*).

The first condition states that both x and y are composed of combinatorial units.⁵ In the Morse system, these are ‘dots’ and ‘dashes’, i.e., short and long physical signals of some sort, whereas in the English system, they are simply letters of the English alphabet. In the DNA and RNA systems, the combinatorial units are the bases, whereas in the protein system, they are the amino acids.

The second condition states that there is a correspondence between two combinatorial systems to which x and y belong. The units of correlation in case of the Morse code are combinations of dots and dashes within the Morse system that can be mapped onto the units of combination within the English system, i.e., the letters, and the latter are at the same time the units of correlation within the English system. In the transcription and replication codes, the units of correlation are simply the same as the units of combination, i.e., DNA or RNA bases. In the translation code, on the other hand, the units of correlation of the RNA system are RNA base triplets or codons, whereas those of the protein system are amino acids.

According to the second condition, there should also be a one–one or many–one mapping between the encoding and the encoded system, i.e., to one correlational unit within the former there is bound to correspond one and only one correlational unit within the latter, but the converse need not hold. Thus, in the case of the Morse code, there is a strict one–one mapping, and the same applies to the transcription code between DNA and primary mRNA, as well as to the replication code – but not to the translation code, as we have seen.

As there is no one–one mapping in the case of the translation code, the syntax of each correlated system is determined in a more complicated way than in the transcription and replication codes: firstly, each correlational unit consists of precisely three combinatorial units, and, secondly, a full chain of amino acids is encoded by a string of codons that begins with a ‘start’ codon and ends with a ‘stop’ codon – there are no syntactical markers beyond this. Since there are 20 canonical amino acids, it follows that the number of correlational units within the encoding system is bound to exceed that of correlational units within the encoded system. Indeed, there are 4^3 possible codons, one of which is a ‘start’ codon and three ‘stop’ codons. As the start codon also specifies an amino acid, namely, methionine, this leaves 61 codons for amino acids. There is thus redundancy on the part of the codons – most extensively in the cases of leucine, arginine and serine, which are each assigned to as many as six different codons. Furthermore, the parsing of RNA syntax into correlational units is of necessity sensitive to the addition or deletion of combinatorial units, unlike that of the Morse syntax. A complete reparsing into different codons or ‘words’ is therefore possible if bases or ‘letters’ are added or deleted – a so-

⁵Since the order of arrangement matters, the combinations in question are of course not combinations in a strict mathematical sense but rather *permutations* (with repetition).

called ‘frame shift mutation’. There is nothing analogous to this with regard to the Morse code, since this code has separate markers of the syntactical units that can be mapped onto English words – the Morse ‘words’ – in the form of longer pauses than those between the correlational units, which can be mapped onto English letters.

The third condition states that the two structures display the same configuration of correlational units, i.e., the position of a correlational unit within x should be the same as that of the corresponding correlational unit within y . The second and third conditions taken together thus assure us that there is a discrete congruence, as opposed to a continuous one.

The fourth condition is there to rule out the possibility that x and y could encode each other, or, at least, if x and y are types, that two *tokens* of x and y could encode each other, as there may be a code of complementarity or even of identity between types.

Is there a genetic code for phenotypes?

We have seen that there is a genetic code for proteins, but often one speaks as if there is also a genetic code for whole phenotypes, i.e., as if genes are part of two different homonymous encoding systems.⁶ Now, if genes carry semantic information by way of the gene–protein code, this raises the question whether there could also be semantic information that is carried by a gene–phenotype code, which, even if weaker than the former, would still be sufficiently similar. I shall therefore discuss whether there is any resemblance between the causal path from genes to proteins and the causal route from genes to phenotypic traits.

There is of course no combinatorial system underlying phenotypic variation in the way there is a combinatorial system underlying protein variation. Thus, if genes are supposed to encode phenotypes, this can at most be in the weak sense that there is a one–one or many–one correspondence as well as an asymmetrical causal relation between genes and phenotypes (cf. conditions 2 and 4 respectively of the correspondence code definition), but no combinatorial discreteness (cf. condition 1), and no pattern congruence (cf. condition 3). Let us call such a weak correspondence code simply a ‘w-code’. Clearly, it is only because a w-code resembles a strong correspondence code that it can be called a ‘code’ at all, which is why I began my discussion with the more fundamental gene–protein code.

A w-code for phenotypic variants that range over a continuum would have to be continuous, like that between temperature and the states of a thermometer. Hence, as genes vary discretely, there can be no w-code between

⁶Cf. the distinction in Moss (2004) between the “epigenetic” gene-D, which is molecular, and the “preformationist” gene-P, which is not molecular. Moss argues that there is a tendency to confuse the two.

genes and continuous phenotypic variants. This should of course not be confused with the claim that discrete differences between genes are incompatible with continuous phenotypic variation, as the genes may interact with variation in other factors and there will always be 'developmental noise'. But a w-code requires a code-like fixation of causal background conditions as well as of the causal pathway, and thus a genetic continuum that is isomorphic with a phenotypic one.

One may think that there are plenty of phenotypes that are not continuous, but quite often those which are modelled as discrete turn out not to be so after all. For instance, elementary textbooks still present Davenport's initial analysis of the heredity of eye colour from 1907 as a paradigmatic example of a monogenic trait determined by two Mendelian alleles, one for brown and one for blue eye colour, where the former is dominant and the latter recessive. In reality, eye colour in Europeans varies over a continuum from light blue to dark brown, including grey, green and yellow. Hence, it cannot correspond to the discrete variation between three different pairs of Mendelian alleles.

Additionally, even if a specific molecular sequence were to be identified with a Mendelian gene for brown eye colour, in that it would determine the melanin pigmentation of the iris, there would not be any specific molecular sequence that would correspond to the Mendelian gene for blue eye colour, i.e., that would w-encode blue eye pigmentation. It is rather that when a molecular sequence for brown eyes is lacking, the iris will not be sufficiently pigmented to absorb much light, and the Rayleigh scattering of the reflected light by the collagen fibrils in the eye will then make it appear blue. There is not even a specific molecular sequence that can be identified with a Mendelian gene for brown eye colour, since it is a polygenic trait.⁷ Indeed, it is precisely because there is no w-code between genes and eye colour phenotypes that there can be continuous variation in the latter.

It thus appears to be a general problem with code talk about traits that there is nothing that can rule out a potential continuum of intermediate cases between two specific phenotypes. Still, for the sake of argument, we may assume that there are some specific phenotypes that are discrete, like sex in mammals, even if we know that there could be hermaphroditic intermediaries in practice. But such phenotypes can only be sharply individuated on the basis of discrete differences between genomes – possibly recognizing a third 'genetic sex', like that of humans with Klinefelter's syndrome, who have both two *x* chromosomes as well as a *y* chromosome. As the phenotypic discreteness cannot be sharply determined independently of the discrete variation of genes themselves, it is a *superimposed* discreteness.

It is clear, then, that there are not in general any genotype–phenotype assignments in the sense in which there are codon–amino acid assignments. The "code talk" about phenotypes probably stems from the fact that there is coded information as well as information about phenotypic traits in the genes. But it

⁷Cf. Sturm and Frudakis (2004).

does not follow from the fact that genes code for proteins, and, thus, that there is coded information even about phenotypes in the genes, that they also code or w-code for phenotypes. There is after all an interaction between genes that determine any polygenic character, and many genes are pleiotropic, which means that there can at most be a w-code between *clusters* of genes and *clusters* of phenotypes. This is of course what is to be expected if genes are just causal factors on a par with other causal factors, since causes and effects in general branch out into trees – there is not just a single mono-linear chain of events.

Despite these shortcomings of a genotype–phenotype semiotics, some may argue that there is at least a w-code between *differences* between genes and *differences* between phenotypes in a *fixed* environment. There are two major problems with this proposal in the present context. Firstly, it is not clear what a fixed environment would amount to. It may not be sufficient to keep the outer environment of a living system constant, nor the rest of the genome, since the process of development is also influenced by other internal conditions, and there is developmental noise. Of course, some developmental processes are more context sensitive than others, but the concept of a w-code between genetic and phenotypic differences in general would seem to require well-defined causal background conditions in general, just as the concept of a code between genes and proteins. However, the complexities of interactions at this level do not seem to permit anything similar to a ‘translation machinery’, except in a statistical sense, even when this is taken to be variable rather than constant – in accordance with the so-called *norms of reaction* in question.

Secondly, even if we assume for the sake of argument that there is a well-defined set of causal background factors that are to be fixed in a particular state, one might just as well argue that there is a w-code between differences between environmental factors and differences between phenotypes when the genome is fixed. To be sure, there is no combinatorial system that accounts for variation in environmental factors, the way there is one that accounts for variation in genes, whereas the reason why genes are said to w-encode phenotypes seems to be that genes code for proteins. However, the inference from a genetic code for proteins to a w-code for phenotypes is invalid, anyway. On the face of it, environmental factors, like sunlight or nutrients, should enable a w-code for phenotypic properties to a much larger extent than genes, since they are continuous in their quantitative dimension. It would seem, then, that there could be an analogue w-code between variation in such factors and that of continuous phenotypes, e.g., between dosages of sunlight or nutrients and the height of plants. Indeed, there can even be a w-code between variation in continuous environmental factors and quasi-discrete phenotypes – like sex. Thus, it is not genetically determined from the outset whether crocodiles will be male or female – it is rather the incubation temperature that determines the sex of the hatchlings, i.e., that “switches on” one set of genes rather than another.

There is still a candidate for a kind of genes which can be said to fulfil a w-code criterion with regard to phenotypes as such without superimposed discreteness, namely, the homeobox genes. Thus, by inducing the expression of

the Pax-6 mouse gene in a *Drosophila* fruitfly, Gehring managed to produce a fly eye on its leg.⁸ Apparently, the same gene that produces a mouse eye in the context of mouse eye genes triggers development of a compound fly eye in the context of fly eye genes, so it seems that at least such homeobox genes can be said to w-encode the development of species-specific phenotypes, like mouse and fly eyes. Indeed, it has been argued that the Pax-6 gene contains the instruction ‘make an eye here!’⁹, a bit similar to the way genes in general might be taken to contain the instruction ‘make a protein!’ It does not follow from this, though, that homeobox genes contain information in a different sense from that in which causes in general contain information about their effects. The reason why the homeobox gene is special is rather that it is a ‘meta-gene’, as it were, controlling the phenotypic expression of other genes, and the discrete difference between it and its mutants can therefore be mapped onto the binary opposition between eye and eyeless phenotypes at a general level of expression. When it comes to specific eye phenotypes, like that of the mouse and the fly, there is no w-code, because of the continuous variation of those. Consequently, most genes can of course still only w-encode by way of superimposed discreteness – if at all, as most genes are not meta-genes.

It would seem, then, that only homeobox genes can be said to w-encode phenotypes that are not just differentially determined, whereas step-wise genetic variation in general can at most be taken to w-encode discrete differences in species-specific and individual phenotypic variation by way of superimposition, i.e., the discreteness of the phenotypic variants is parasitic on that of genetic variants and cannot be individuated independently – unlike whole organs or organelles as such, which are individuated functionally. Since, by parity of reasoning, variation in environmental factors can also be said to w-encode continuous, and even quasi-discrete phenotypic variation, a w-code does not distinguish between genetic and environmental factors, and therefore it does not provide genes with any special informational role to play either.

The upshot of this brief analysis is that in no way can an analogue w-code or a digital w-code with superimposed discreteness or mere meta-discreteness be assimilated to the genetic code for proteins. Although w-codes amount to causal co-variation, they do not consist in intersystematic correlation based on intrasystematic combinations, as does the Morse–English code. It is combinatorial discreteness as opposed to superimposed or meta-discreteness and combinatorial correspondence as opposed to cluster correspondence that is the key to understanding the genetic code as a genuine code, then, and therefore as possibly carrying information in a different sense from that in which causes in general can be said to carry information about their effects. We are thus left with the idea of DNA coding for DNA, for RNA as well as for proteins, i.e., with the replication, transcription and translation codes, and we shall now return to the discussion of these codes.

⁸Cf. Callaerts et al. (1997).

⁹This is how it is interpreted in Maynard Smith (2000a, p. 188f).

The arbitrariness of the genetic code

The translation code is often claimed to be *arbitrary*, as opposed to the replication and transcription codes, and we shall now look into what this arbitrariness might amount to.

That the codon–amino acid assignments are arbitrary might simply mean that they *could* have been different even if the physical laws and chemical principles were the same, i.e., that they are not physico-chemically necessary. But this would not distinguish the relation between codons and amino acids from causal relations in general, as there can always be a change in the background conditions of a causal relation, and we do not in general say that contingent relations are arbitrary.

That the codon–amino acid assignments are arbitrary could also mean that they are due to natural selection, and thus not merely physico-chemically contingent, but that they were actually selected over alternative assignments. After all, why speak of ‘assignments’ in the first place if it were just a matter of counterfactual-supporting co-variation? One problem with this is that the translation code may be due to a frozen accident rather than to selection, as has been argued by Crick.¹⁰ Hence, the arbitrariness of the genetic code cannot simply be attributed to its selection history. To be sure, Crick’s hypothesis has come under attack recently, but there is no indication that the code derives from natural selection in the same strong way as adaptations do. Rather, it might be the outcome of a complex interplay between chance, selection and stereochemical factors.¹¹ Moreover, even if the translation code were due to natural selection, this would not distinguish these assignments from anything else for which there has been natural selection. After all, whenever there is an adaptation, we can imagine alternative routes that the process of evolution could have taken – if the initial conditions had been different, or if the environment had been different, or if the spontaneous mutations had been different – possibly leading to functional equivalents. If it is objected that it is only with regard to a code that it makes sense to speak of arbitrary assignments, the problem is that selection would not distinguish the translation code from the transcription and replication codes either, as there could have been selection for alternative transcription and replication codes as well.

There is another sense of ‘arbitrary’ that applies to codon–amino acid assignments that is more interesting, in that it actually does bring out a difference between the replication and transcription codes on the one hand and the translation code on the other. Thus, it is easy to realize that the replication code is based on mere complementarity between the DNA bases of each strand of the original molecule and those that are chained together to form the new strand of each of the daughter molecules. The transcription code between DNA base triplets and mRNA codons (with uracil substituted for thymine) is

¹⁰Cf. Crick (1968).

¹¹Cf. Knight et al. (1999).

also based on complementarity, namely, the one between anti-sense DNA and mRNA, which mirrors the replication code between sense and anti-sense DNA. Sense DNA is then said to encode amino acids even when it is actually the anti-sense DNA that is the cause of the mRNA transcript. However, this should just be taken as notational shorthand for the fact that each RNA codon is caused by a corresponding DNA base triplet.

The further translation process maps mRNA codons onto amino acids by way of complementary tRNA anticodons, each of which is attached to a specific amino acid. Admittedly, the complementarity only applies to two of the three bases in the codons, in that guanine in the third base is ‘wobbly’, and can be paired not only with cytosine but also with uracil, so that one and the same tRNA anticodon may correspond to two different mRNA codons. This helps explain the degeneracy of the genetic code. Indeed, in many cases all four mRNA codons that only differ in their third base code for the same amino acid, thus constituting a ‘family box’. Still, there is at least a partial complementarity between sense DNA and tRNA anticodons, and thus an overall causal chain of physico-chemical interactions which seems to justify an extension of the transcription metaphor to cover also the mapping of mRNA codons onto tRNA anticodons – at least within the idealized version of the process that ignores alternative splicing and other complexities.

However, the attachment of amino acids to tRNA anticodons is mediated by 21 aminoacyl-tRNA synthetase enzymes that enable the aminoacylation (there are two for lysine). It is the relation between the amino acid activation site and the tRNA binding domain of each of these enzymes that determines which amino acid a codon specifies. It is therefore possible to change such an enzyme or even to engineer completely new enzymes so that *some but not necessarily all* of the codons will specify different amino acids. There are no physical laws or chemical principles that rule out such an “interchange” of the amino acid activation sites of the enzymes.¹² This distinguishes the translation code from the replication and transcription codes. Of course, it is also possible to imagine alternative transcription and replication codes, but only in the sense that the entire codes could have been based on a different principle than that of sequence complementarity. So even though there are enzymes that take part in the processes of replication and transcription – like helicases and polymerases – these are not *assignment* enzymes that determine the specific correspondences of the codes.

In the translation code, on the other hand, there can be a change in *specific* correlations while other correlations remain unaltered, just as there can be a change in individual assignment rules in a Morse code, without thereby setting up a completely different code. The enzymes in question are thus the closest we get to particular biological assignment rules, and their recognition-specificity is arbitrary in a similar way to that in which the assignments of a man-made code are arbitrary. Hence, although the process that leads from DNA to tRNA is

¹²For an analysis of ‘chemical necessity’, see Stegmann (2004).

not arbitrary in this sense, the translation code *is*, and the entire correspondence between DNA base triplets and amino acids – the genetic code within the idealized model of protein synthesis – is therefore arbitrary in the relevant sense.

It will be recalled, though, that the genetic code is based on a discrete pattern congruence, so a gene can only encode the sequence of amino acids in a protein, and not its further non-combinatorial structure. Hence, it can encode its primary structure, or that of one of its polypeptide chains – if it has several, i.e., the linear sequence of amino acids, but not the three-dimensional coiled, parallel or folded arrangement of the molecules. Still, the latter, secondary and tertiary structure is very important with regard to the chemical properties of a single-chained protein or polypeptide subunit, and the same applies to the quaternary structure of a multi-chained protein. But these higher level structures follow from the laws of physics and chemical principles when the primary structure is given, whereas the primary structure itself does not follow from these when the sequence of base triplets or codons is given. At least, it cannot do so if the codon–amino acid assignments really are arbitrary.

To expand on this, I shall introduce a distinction between the ‘digital’ and the ‘analogue’ structure of a molecule, or of a non-branching molecular subunit. By the former, I shall understand its discrete, combinatorial structure, i.e., a linear sequence of molecular subunits, and by the latter, its non-combinatorial structure. The amino acid sequence that makes up the primary structure of a protein – or of a polypeptide subunit – is thus the digital protein structure, as opposed to its higher order analogue structure of coils and folds. Even the DNA and RNA molecules of course have an analogue structure – which is usually a double or simple helix, respectively – but sequences of DNA base triplets or RNA codons do not. Whenever we speak of a genetic code for proteins, we should bear in mind that it is just a code between the digital structure of DNA or RNA on the one hand and the digital structure of proteins on the other, i.e., between base sequences and amino acid sequences – there is no code between the three-dimensional shapes of full-fledged molecules. Still, as we shall see, the analogue structure of a protein is very important with regard to its informational role.

The arbitrariness of the correlation of the digital structure of DNA with that of proteins, then, is one that derives from the arbitrariness of the assignment of amino acids to codons – a bit similar to the way the arbitrariness of a Morse word or sentence derives from that of the correlational units of the Morse–English code. However, it is not clear that this arbitrariness is *essential* to the difference between the replication and transcription codes on the one hand and the translation code on the other. Even if there were no such arbitrariness, one might argue that there would be an arbitrary relation between genes and the function of the proteins they encode, in the sense that there is no pattern congruence between them. Functions do not constitute a combinatorial system, so there can be no correspondence between a combinatorial DNA system and a combinatorial system of amino acid functions, i.e., there is no discrete

congruence between a sequence of codons and a sequence of functional subunits of a protein. Indeed, it does not make much sense to speak of simple functions as subunits of complex functions at the protein level. Genes can therefore not be said to encode functions in the same strict sense as they encode proteins.

It follows from all this that even if the codon–amino acid assignments were not arbitrary, the relation between the proteins and their functions could still be said to emerge as arbitrary in the sense that it would not be determined combinatorially on the basis of amino acids functions. Hence, the gene would not determine the protein function in such a combinatorial manner either. This is clearly a code-dependent notion of arbitrariness – in that it is only because proteins are code-patterned – thereby also conserving the code pattern of genes – that protein functions can be said to be arbitrary in that they lack such a pattern. There is therefore no way in which new gain-of-function and loss-of-function mutations can be predicted on the basis of changes in the digital gene or protein syntax alone. Indeed, the function of a protein depends on its analogue structure, but it is only the digital one that is genetically encoded. The very relation between the digital and the analogue structure of a protein is therefore the source of a new form of arbitrariness that pertains to the protein functions, and which I shall call *emergent*, as opposed to the *assignment* arbitrariness of the codon–amino acid encodings and the resultant, correlational one of whole gene–protein encodings that derives from the latter.

The definition does not entail that all functions display emergent arbitrariness – only functions of objects that take part in a combinatorial system, like the protein system, do. Functions of phenotypes are thus not arbitrary according to this definition. Nor are DNA or RNA molecules, even if they take part in a combinatorial system, since they do not in general have functions – in the sense of causal roles beyond those of being replicated, transcribed and translated.

The question is now whether either correlational arbitrariness or emergent arbitrariness is a sufficient reason to regard the causal process that leads from codons to amino acids as a *translation*, as opposed to the ‘replication’ process that leads from DNA to DNA, or the ‘transcription’ process that leads from DNA to RNA. To answer this question, we have to analyze the analogy between the translation of genes into proteins and that of Morse into English further.

From genetic code-script to semantic content?

The translation from Morse into English is only a quasi-translation, since a genuine translation is between a source language and a target language that have independent systemic meaning, and not simply a substitution deciphering into one such system. The semantic content at type level thus stems from the

target language alone, i.e., from the English language as a system of norms of interpretation, and not from the Morse quasi-language.

Now, the most pressing question within our analysis of genetic information is whether the resemblance between the Morse–English code and the gene–protein code is sufficient to ascribe a semantic content to genes as well as to proteins, i.e., if there is a protein language that can play a role analogous to that of an English target language. Indeed, *semantic information may be said to be ‘transmitted’ or ‘transferred’ from a token for a gene type x to a token for a protein type y by way of a genetic code if and only if the following conditions are fulfilled: (1) x encodes y ; (2) y has semantic content; (3) there is causal co-variation between the token for x and the token for y .* While it is clear that conditions 1 and 3 are fulfilled on the basis of the definition of a correspondence code, it is far from clear that condition 2 is fulfilled. Whether it is or not, will be the topic of the rest of this paper. If it is not fulfilled, there can be no semantic content in genes by way of an analogy with the Morse–English code either.

To settle the issue of whether genes and proteins have semantic content or not, we have to decide whether they are sufficiently similar to the words and sentences of a man-made code. I have introduced code-dependent notions of arbitrariness for precisely this purpose. My construal of different types of arbitrariness in terms of linguistic code resemblance, so as to justify semantic relevance, thus extends way beyond the restrictive view of Maynard Smith, who maintains that while regulatory proteins and hormones are symbols, enzymes are not.¹³ In my view, all kinds of proteins can be said to be symbols in that their amino acids make up a combinatorial system – like the letters of the English language – and they display emergent arbitrariness with regard to their functions – as English words do with regard to the semantic content that emerges when their letters are put together. Indeed, if genes are to be symbols, like Morse words, and translated into the digital structure of proteins, i.e., amino acid chains, then proteins must also be symbolic, since symbols can only be translated into symbols. Regulatory proteins just have a more complex way of symbolizing their functions than other proteins.

Accordingly, as long there is correlational arbitrariness, and thus conventional determination of protein functions in a naturalized sense, genes for proteins may be taken to parallel man-made symbols of symbols, like Morse words symbolizing English words. And if there were no correlational arbitrariness between genes and proteins, there would still be emergent arbitrariness, and thus both gene as well as protein symbols of protein functions. As genes are also part of a combinatorial system, I think the remoteness of the functions of the proteins they encode from the structure of the genes themselves would suggest that it is *as if* they are conventionally determined even without assignment arbitrariness. The importance of molecular assignment arbitrariness

¹³Cf. Maynard Smith (2000a, p. 193).

ness and the correlational one between genes and proteins that flows from it is thereby somewhat downplayed.

It follows that a continuous congruence between two structures, like that between a map and a landscape, can never be symbolic, despite the fact that there are conventions involved with regard to how the map represents the landscape. The continuous congruence is simply too much of a resemblance, and there are no units of combination and correlation, and thus neither assignment arbitrariness nor emergent arbitrariness. Ironically, a gene is often conceived of as a template or blueprint for the proteins, even if this connotes iconic resemblance rather than symbolic arbitrariness. In reality, it is precisely because a DNA gene is not literally a template or a blueprint that it can represent symbolically.

Furthermore, in the ‘no DNA’-world envisaged by Godfrey-Smith,¹⁴ where proteins “act as a template” for their own replication – by way of chemical affinities and ‘connector molecules’ – there is no correlational arbitrariness, but the ‘protein genes’ can still be said ‘to code for themselves’ in terms of an *iconic* relation, and they are still symbols in terms of the emergent arbitrariness of their functions. This protein–protein code would thus be analogous to an English–English code, so to speak, as opposed to a Morse–English code. This again confirms the importance of emergent arbitrariness as opposed to correlational arbitrariness, as the latter may be evolutionarily contingent, unlike the former.

The analogy between the emergent arbitrariness of protein functions and that of the semantic content of English words would suggest that it is the protein functions themselves that have to be identified with semantic content. In so far as the DNA or RNA sequences do not have functions, this distinguishes the translation code from the replication and transcription codes. We know, however, that a very high percentage of DNA does not code for proteins. It may be a regulatory sequence, like that of a promoter region. Or it may be so-called ‘junk DNA’, which either lies between genes or within them. It has recently been discovered that much of this *intergenic* and *intronic* DNA may code for RNA.¹⁵ In particular, many regions of RNA corresponding to introns contain microRNAs that can regulate other genes directly – so-called RNA-interference, without any protein intermediary, unlike, say, gene repression by regulatory genes. This means that it is not proteins but the active RNA itself in this case which has microfunctions. As there is a combinatorial RNA system, these functions also display emergent arbitrariness, but there is of course no correlational arbitrariness between intron-DNA and active RNA. Arguably, the significance of the standard definition of a molecular gene is thereby somewhat diminished.¹⁶ Still, I shall concentrate on the translation code, since

¹⁴Cf. Godfrey-Smith (2000a).

¹⁵Cf. Eddy (2001) and Mattick (2003).

¹⁶To complicate matters further, there are also ‘riboswitches’, which is a form of RNA that does not code for a protein as such, but which can still cause one in response to a particular chemical agent. Cf. Vitreschak et al. (2004).

if functions of proteins cannot be semantic, there seems to be no good reason why functions of DNA or RNA should be semantic either.

We have seen, then, that the genetic code is relevantly similar to the Morse code with regard to arbitrariness. However, a further property of linguistic sentences is that they can be true or false, and what remains to be shown is that the gene–protein code is also analogous with regard to this representational aspect. Clearly, arbitrariness alone will not be able to take care of it. Rather, *aboutness* – or *intentionality* – is required as well, just as in the case of a man-made code. Indeed, it is often held that if there cannot be misrepresentation there cannot be representation either.¹⁷ Is there anything corresponding to misrepresentation with regard to the gene–protein code? Perhaps if a protein is dysfunctional – as when it is no longer adaptive, which should of course not be confused with the possibility of an error in the developmental process that is “supposed to” – in a naturalized sense – lead from gene to protein function, as that would rather be misinterpretation. We thus have to look into the relations between genes and protein functions to determine whether there is semantic information content that can be misinterpreted through development or even misrepresent its own developmental outcome.

Whereas aboutness derives from a human coder within Morse code-script, that of genetic code-script can only stem from natural selection. Thus, Maynard Smith argues that in biology “the coder is natural selection. (...) It is natural selection which, in the past, produced the sequence of bases, out of many possible sequences, which, via the information channel just described, specifies a protein that has a ‘meaning’, in the sense of functioning in a way that favors the survival of the organism” (Maynard Smith 2000a, p. 179). Consequently, while the relation between digital and analogue structure is the source of the emergent arbitrariness that can possibly be identified with that of semantic content, DNA, RNA and proteins would have such content only in so far as they possess their digital structure because of natural selection.

However, if natural selection is to be the coder, a gain-of-function mutation cannot possess semantic content on its first occurrence in a germ-line – nor on its repeated occurrence within the soma before selection. Hence, a widespread mutation may suddenly lack semantic content if it is not inherited but appears locally new to natural selection. Furthermore, even repeated occurrences of a mutated gene within one and the same lineage cannot possess semantic content if it is due to random genetic drift – which may be particularly important at a molecular level – or merely selected because it is linked with other genes for which there is selection.¹⁸

¹⁷According to the version of Shannon’s model of communication presented in Dretske (1981), even causal information (or possibly quasi-causal, if indeterministic processes are not considered to be causal) is semantic. This is however linked with a reliabilist theory of knowledge, so it yields no basis for semantic information in the more narrow sense that we are after.

¹⁸Cf. Sober’s distinction between selection *of* and selection *for* (Sober 1984). It is only the latter that can bring about teleosemantic information. Cf. also Dennett (1987, p. 316) on intentionality and selection *for*.

This of course makes a gene or a protein very different from a man-made word or sentence, which has a semantic content at type level, as it is only at token level that selection can make a difference between a ‘causal role’ function and an ‘etiological’ function, and thus between causal and semantic information. In short, while gene and protein *tokens* may fulfil both the arbitrariness and the aboutness conditions of a strong criterion of semantic content, gene and protein *types* will fulfil only the former.

That there is no aboutness at type-level is also borne out by the fact that the function of a protein may vary from one cell to another within the same organism. Thus, ‘lactate dehydrogenase’ and ‘crystallin’ name the same molecule, but the former denotes it in so far as it is an enzyme in muscle cells, while the latter applies to it qua structural protein in the eye lens. It seems that in principle all proteins can play causal roles that are so widely different that they cannot possibly be semantically related, i.e., there is no token-invariant manner of representation that can be attributed to the protein type.

It follows that *there can be no ascription of semantic content to a gene or protein type that can be shared indiscriminately by its tokens*. But it was the very analogy between the genetic code and the Morse code that motivated the strong criterion of semantic content in the first place – i.e., the idea that genes and proteins carry information in a language-like manner. As the criterion cannot be applied at type-level but only to a proper subset of gene and protein tokens, it cannot justify the claim that the genetic code is similar to the Morse code in a semantically relevant manner, i.e., in terms of the *kind* of information that is carried by genes and proteins. Nor does there seem to be any weaker, code-independent criterion that would yield such a distinction with regard to genetic information.

What is peculiar to genes and proteins is rather *how* they carry their causal information, namely, in terms of not only correlational but also emergent arbitrariness – with the scope of functional variation that is thereby enabled by alterations in gene- and protein code-script. To be sure, to a human interpreter, the discrete pattern congruence of the genetic code may still seem to resemble a transparent man-made arrangement, and thus to be semantically significant. But this transparency is deceptive if it leads to semantic content ascriptions. The mindless natural processes are blind to it – it is an epistemic notion rather than an ontological one – and it can play no part in the causal processes of nature. At most, the digital, as opposed to analogue, character of the codes brings about higher fidelity in the processes in which they are involved. But high fidelity is not a characteristic of the transmission of semantic information in particular. It is rather a hallmark of mechanical transmission of information – be it semantic or not.

Acknowledgements

I am grateful to Kim Sterelny and an anonymous referee for comments on an earlier draft of this paper. I have also benefited from discussions with Nils Roll-Hansen.

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